

INPUT

samplesheet.csv

single-end ONT FASTQ (.fastq.gz)



--gene_quant_mode ■ exactly ONE branch runs per execution

epi2me ■ QUANTIFICATION_EPI2ME

Requires: stringtie_enabled=true, genome FASTA + GTF

FILTER_PRIMARY_BAM (-F 0x904)

STRINGTIE ■ per-sample transcriptome GTF

GFFREAD ■ transcriptome FASTA

MINIMAP2 txome align (R2 FASTQ, map-ont)

GFFCOMPARE (StringTie vs reference GTF)

EXTRACT_BAM_TAGS + ASSIGN_FEATURES

CREATE_MATRIX
10x MEX ■ gene + transcript ■ integer UMIs

Output: {outdir}/epi2me/
create_matrix ■ stringtie ■ gffcompare

isoquant ■ QUANTIFICATION_ISOQUANT

Requires: genome FASTA + GTF ■ IsoQuant 3.13

FILTER_PRIMARY_BAM
optional ■ --isoquant_use_primary (default true)

UMITOOLS_DEDUP
--per-cell ■ directional ■ CB/UB

SAMTOOLS_INDEX + SAMTOOLS_FAIDX

ISOQUANT 3.13
genome BAM ■ --barcoded_bam ■ --read_group tag:CB
sensitive_ont ■ unique_only (tx) ■ counts_format mtx

Grouped MTX outputs
gene + transcript ■ per-barcode matrices
+ discovery GTFs ■ read assignments

Output: {outdir}/isoquant/
filter_primary_bam ■ umitools_dedup ■ isoquant tables

oarfish ■ QUANTIFICATION_OARFISH

scnanoseq-style ■ no genome alignment ■ requires GTF + FASTA

GFFREAD + MINIMAP2_INDEX txome
reference transcriptome from GTF

MINIMAP2_ALIGN_TXOME
R2 FASTQ ■ map-ont ■ -N 100

SAMTOOLS_FILTER_MAPPED (-F 4)

TAG_BAM (CB / UB)

UMITOOLS_DEDUP ■ SORT_BAM_CB

OARFISH 0.9.4
--single-cell ■ --filter-group no-filters

OARFISH_AGGREGATE
10x MEX ■ transcript + gene (GTF aggregation)

Output: {outdir}/oarfish/
transcriptome_align ■ quant ■ gene_bc_matrix

LEGEND

Optional step

Shared preprocessing

epi2me branch

isoquant branch

oarfish branch (skips genome align)